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beregninger
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Innlevering 2

1 a) With the given equation,

$$\frac{d^2}{dx^2}y(x) = -4 \sin 2x \quad (1)$$

(2)

integrating twice gives,

(3)

$$\frac{d}{dx}y(x) = 2 \cos 2x + C \quad (4)$$

$$y(x) = \sin 2x + Cx + D \quad (5)$$

With the initial conditions,

$$y(0) = 0, \quad y'(0) = 2 \quad (6)$$

(7)

we get that,

$$y'(0) = 2 \cos(2 \cdot 0) + C = 2 \Rightarrow C = 0 \quad (8)$$

and,

$$y(0) = \sin(2 \cdot 0) + D = 0 \Rightarrow D = 0 \quad (9)$$

$$y(x) = \sin 2x \quad (10)$$

b) Defining the system as,

$$\vec{y}(x) = \begin{pmatrix} y(x) \\ y'(x) \end{pmatrix} = \begin{pmatrix} y_1(x) \\ y_2(x) \end{pmatrix} \quad (11)$$

it can now be written as a first order differential equation,

$$\frac{d}{dx}\vec{y}(x) = \begin{pmatrix} y_2(x) \\ -4 \sin 2x \end{pmatrix} \quad (12)$$

with initial conditions,

$$\vec{y}(0) = \begin{pmatrix} y(0) \\ y'(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad (13)$$

- c)
- d)
- e)
- f)
- g)
- h)

- 2** a) To assure that the functions *matrix_to_vector()* and *vector_to_matrix()* work correctly and are inverse, the test function *test_matrix_vector_conversion()* was made.

The test function creates a 2x2 complex matrix,

$$\begin{pmatrix} 1 + 2i & 3 + 0i \\ -1 - 2i & 0 + 3i \end{pmatrix} \quad (14)$$

and then uses *matrix_to_vector()* to transform this matrix into a 8-component vector. The test then uses the function *vector_to_matrix()* to reconstruct a 2x2 matrix from the 8-component vector.

The reconstructed matrix is then compared to the original matrix (with *np.allclose* to avoid issues with floating-point precision errors, due to how floats are represented on a computer). If the reconstructed matrix is not the same as the original one, the function will print out an error message. If it is, then it will print out the transformation and reverse-transformation was successful.

- b) To assure the functions *four_vectors_to_32_vector()* and *vector_32_to_four_vectors()* work correctly and are inverse, the test function *test_vector_32_conversion()* was made, similar to that of task **2a**.

The test function creates four random 8-component vectors and uses the functions *four_vectors_to_32_vector()* to transform them into a single 32-component vector and uses *vector_32_to_four_vectors()* to reconstruct them into four 8-component vectors. The four reconstructed vectors are then compared to the original random generated ones (with *np.allclose*) and prints an error message if incorrect or a message to indicate that it worked if it was successful.

- c) The transformations executed by the code can algebraically be represented by the following diagram,

$$\begin{array}{ccc} \mathbb{R}^{32} & \xrightarrow{f} & (\mathbb{R}^8)^4 \\ & \searrow & \downarrow g \\ & g \circ f & M_2(\mathbb{C})^4 \end{array}$$

where $(\mathbb{R}^8)^4$ is the space of tuples of 4 vectors in $\mathbb{R}^8$, and $M_2(\mathbb{C})^4$ is the space of tuples of 4 complex 2x2 matrices. The pre-image of $g \circ f$ will consist of vectors

of the form

$$\vec{v} = \begin{pmatrix} a_{r\gamma} & b_{r\gamma} & c_{r\gamma} & d_{r\gamma} & a_{i\gamma} & b_{i\gamma} & c_{i\gamma} & d_{i\gamma} \\ a_{r\tilde{\gamma}} & b_{r\tilde{\gamma}} & c_{r\tilde{\gamma}} & d_{r\tilde{\gamma}} & a_{i\tilde{\gamma}} & b_{i\tilde{\gamma}} & c_{i\tilde{\gamma}} & d_{i\tilde{\gamma}} \\ a_{r\omega} & b_{r\omega} & c_{r\omega} & d_{r\omega} & a_{i\omega} & b_{i\omega} & c_{i\omega} & d_{i\omega} \\ a_{r\tilde{\omega}} & b_{r\tilde{\omega}} & c_{r\tilde{\omega}} & d_{r\tilde{\omega}} & a_{i\tilde{\omega}} & b_{i\tilde{\omega}} & c_{i\tilde{\omega}} & d_{i\tilde{\omega}} \end{pmatrix} \in \mathbb{R}^{32}.$$

This composition is the identity map from a $\mathbb{R}^{32}$ to $M_2(\mathbb{C})^4$, and is executed in the code by the function *decompact_riccati*, and $(g \circ f)^{-1}$ is executed by *compact_riccati* in the code given above.

d)

e) The function *usadel_rhs* returns a (32, m) array, which is the format required by *solve_bvp*. A potential problem is that, in theory, the function should only take in two arguments, one for the x-values we work with, and one for the derivatives of v at each x-coordinate. This problem will be solved later by using a lambda function when calling *solve_bvp* to make the function compatible.

f) The function *boundary_conditions_normal* returns a 32 element vector, which is required later when we want to solve the boundary value problem. However, since the function also depends on l, a lambda function must be used to get the right format when solving later on, as mentioned earlier.

g) We used the lambda functions as described earlier and the result is as physically expected for a normal metal lined with normal metals; the solution is numerically identical to zero.

h) The plots of the DOS for the different values of ε show a constant density at $\frac{D(x,\varepsilon)}{D_0} = 1$. This is again as physically expected for normal metals. This reflects the fact that the number of available electron states per unit energy remains constant in the normal metal.

i)

j) Considering Fig. 1 we observe that the relative density of states is a little above 1 in a superconducting $\varepsilon = 2$. This implies that DOS is a little higher than that of a normal metal, and the DOS in the normal metal in between, will conversely become higher in at the edges due to the proximity effects penetrating into the normal metal. Since ε is relatively high, the effect is rather minor.

k) The plots show DOS in the middle of the normal metal gap as a function of the energy, normalized against the superconducting gap and relative to the Fermi energy, ε , for different gap-lengths.

We observe that the mini gap for small gap lengths gradually decreases as l increases. This is, physically, as expected because of superconducting proximity effects. This meaning, where the superconduction properties penetrate into the normal metal and induce a gap in the density of states in the normal metal. This effect will decrease with the distance from the superconductor itself.

The sharp peaks, which also gradually decrease in magnitude as l increases, occur due to the accumulation of quasi particles at the edges of the mini gap. This happens because states that normally would exist inside the mini gap region are pushed away by the superconducting properties leaking into the normal metal, and the states accumulate just outside the mini gap.

The small peaks just around $\varepsilon \approx 1$ occur due to the wave nature of the quasi particles, and the phenomenon known as Andreev reflection. This happens when a Cooper pair interacts with the interface between the superconductor and the normal metal. The normal region will act as a resonant chamber for the particle-waves with energies close to that of the gap energy. This causes a constructive interference, and thus increases the density of states within the whole normal region. However, we see that this effect diminishes drastically when the length of the gap increases.

- l) We observe that the current integrand is approximately zero. The deviance for some ε is most likely caused by errors regarding the floating point representation of numbers. We conclude that it is numerically identical to zero, which is what is physically expected.

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- m) Here we clearly see that the current integrand is non-zero for some values of epsilon. This is as physically expected, since a phase difference between the superconductors will allow super current to flow through the normal metal under certain conditions. We observe some key features from the plot. The current integrand starts at zero for $\varepsilon = 0$, and decreases until it reaches a maximum in absolute value at around $\varepsilon \approx 0.25$. This corresponds to the energy where the edge of the mini gap is situated in the middle of the metal for the given length $l = 1$ (See the second plot in 2k). The current decreases in absolute value as ε increases from this point, until the resonant effects occurring around $\varepsilon \approx 1$ kicks in. This causes a spike in the current integrand's absolute value, before it shortly jumps up and changes direction when the energy relative to the Fermi energy, $E - E_f$, becomes greater than the energy scale of the superconducting gap $|\Delta|$. The resonant effects quickly diminish as the energy increases from here, and the system loses its ability to conduct super current.

Moving on to the other question posed by the exercise, we solved this exercise before the change in the exercise sheet, which was posted only one day before the due date and wasn't made aware of this until in the evening, the day of the deadline. We have therefore used a different method than what the note implied should be used. So, to check theoretically, whether or not the current integrand is conserved as a function of position, we evaluated $\partial_x j(x, \varepsilon)$.

Starting from the definition,

$$j(x, \varepsilon) = \Re \{ \text{Tr} [\hat{\rho}_3 (\hat{g}(x, \varepsilon) \partial_x \hat{g}(x, \varepsilon) - (\partial_x \hat{g}(x, \varepsilon)) \hat{g}(x, \varepsilon))] \} . \quad (15)$$

taking the derivative, using the fact that the derivative and the matrix trace

both are linear, and not denoting the variables in the functions, we get,

$$\partial_x j(x, \varepsilon) = \partial_x \left(\Re \left\{ \text{Tr} \left[\hat{\rho}_3 \left(\hat{g} \partial_x \hat{g} - (\partial_x \hat{g}) \hat{g} \right) \right] \right\} \right) \quad (16)$$

$$= \Re \left\{ \text{Tr} \left[\partial_x \left(\hat{\rho}_3 \left(\hat{g} \partial_x \hat{g} - (\partial_x \hat{g}) \hat{g} \right) \right) \right] \right\} \quad (17)$$

$$= \Re \left\{ \text{Tr} \left[\hat{\rho}_3 \left((\partial_x \hat{g})^2 + \hat{g} \partial_x^2 \hat{g} - (\partial_x^2 \hat{g}) \hat{g} - (\partial_x \hat{g})^2 \right) \right] \right\} \quad (18)$$

$$= \Re \left\{ \text{Tr} \left[\hat{\rho}_3 \left(\hat{g} \partial_x^2 \hat{g} - (\partial_x^2 \hat{g}) \hat{g} \right) \right] \right\} \quad (19)$$

$$= \Re \left\{ \text{Tr} \left[\left(\hat{\rho}_3 \hat{g} \partial_x^2 \hat{g} - \hat{\rho}_3 (\partial_x^2 \hat{g}) \hat{g} \right) \right] \right\} \quad (20)$$

$$= \Re \left\{ \text{Tr} \left[\hat{\rho}_3 \hat{g} \partial_x^2 \hat{g} \right] - \text{Tr} \left[\hat{\rho}_3 (\partial_x^2 \hat{g}) \hat{g} \right] \right\} \quad (21)$$

$$= \Re \left\{ \text{Tr} \left[\hat{\rho}_3 \hat{g} \partial_x^2 \hat{g} \right] - \text{Tr} \left[(\partial_x^2 \hat{g}) \hat{\rho}_3 \hat{g} \right] \right\} \quad (22)$$

$$= \Re \left\{ \text{Tr} \left[\hat{\rho}_3 \hat{g} \partial_x^2 \hat{g} \right] - \text{Tr} \left[\hat{\rho}_3 (\partial_x^2 \hat{g}) \hat{g} \right] \right\} \quad (23)$$

$$= \Re \left\{ \text{Tr} \left[\hat{\rho}_3 \hat{g} \partial_x^2 \hat{g} - (\partial_x^2 \hat{g}) \hat{\rho}_3 \hat{g} \right] \right\} \quad (24)$$

$$= \Re \left\{ \text{Tr} \left[[\hat{\rho}_3 \hat{g}, \partial_x^2 \hat{g}] \right] \right\} = 0 \quad (25)$$

where we made use of the cyclic property of the matrix trace, and the fact that the trace of a commutator is always zero.

- n)** In the plot, we observe a sinusoidal 2π -periodic relationship between the current and the phase difference. This is known to be a fundamental property of the supercurrent through a normal region due to the Josephson effect!